

WOOSEOK KIM

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Ottawa, ON, Canada

EDUCATION

- Carleton University; Ottawa, Canada.** September 2025 – August 2027
M.S. Computer Science (Data Science Specialization)
- San Jose State University; San Jose, California, United States.** July 2024 – August 2024
Certificate of Achievement: Silicon Valley Software Technology & Artificial Intelligence Program
- Sangmyung University; Seoul, Republic of Korea.** March 2019 – February 2025
B.S. Computer Science GPA: 97.0 / 100

RESEARCH EXPERIENCE

- Research Assistant** September 2025 – Present
Intelligent Machines Lab (iML), Carleton University
- Advisor: Professor Majid Komeili
- Undergraduate Research Intern** June 2023 – December 2024
Network Information Systems Lab, Sangmyung University
- Advisor: Professor KyungSeop Shin
 - First-author of a SCOPUS-indexed journal paper, titled "Optimal Power Allocation and Sub-optimal Channel Assignment for Downlink NOMA System Using Deep Reinforcement Learning".
 - Conducted comparative analysis of Bayesian uncertainty quantification methods on NASA's CMAPSS turbofan degradation dataset, evaluating prediction reliability under sensor noise.
 - Designed and implemented policy-based motion planning for an autonomous vehicle simulator using Deep Reinforcement Learning, and led technical seminars for lab members.

RESEARCH PROJECTS

- Optimal Communication Resource Allocation Using DRL** January 2024 – March 2025
Sangmyung University
- Proposed a deep reinforcement learning framework integrating the REINFORCE algorithm for joint resource allocation (JRA) in downlink NOMA systems.
 - Identified limitations of the existing JRA method, particularly their poor performance in channel assignment under dynamic conditions.
 - Conducted extensive simulation experiments to evaluate the effects of the environment size, neural model architecture, and hyperparameters on convergence behaviour and performance metrics.
 - Demonstrated a 39% improvement in resource allocation performance using an on-policy policy-gradient approach.
- Brain Tumor Detection Using PK-YOLO** September 2025 – December 2025
Carleton University
- Improved PK-YOLO in brain tumor detection by outperforming state-of-the-art performance in axial plane data by 6.4% in recall and 3.8% in mAP@50.
 - Applied a Mixture-of-Experts framework with plane-specific backbone models to mitigate performance degradation across MRI planes.
- Dynamic Soft Prompting Using LION** October 2025 – December 2025
Carleton University
- Extended the LION MLLM by proposing a dynamic soft prompting mechanism to mitigate hallucinations in ViT tasks.

- Achieved VQA performance gains by 4.4% over LION, validating the effectiveness of dynamic prompting.
- Fine-tuned a BERT tag encoder using contrastive learning via parameter-efficient-fine-tuning (PEFT).

Battery Analysis Using Dynaformer

March 2024 – May 2024

Sangmyung University

- Implemented a Transformer-based Dynaformer model to predict lithium-ion battery voltage.
- Trained and evaluated on real-world datasets, achieving strong performance comparable to state-of-the-art performance.
- Analyzed the encoder representations, and captured battery aging and discharge characteristics via PCA.

PROJECTS

Ongoing Projects

- Designing an AlphaZero-style Reinforcement Learning framework for the game of "Gomoku" (Five in a Row). Investigating the use of attention mechanisms to incorporate human-defined patterns by computing attention between the pre-defined patterns and the current board state.
- Analyzing the relationship between ambient light levels and crime rates using Spearman's rank correlation. Furthermore, examining feature extraction to avoid poor feature representation driven by inaccurate dataset and to employ feature vectors.

Reinforcement Learning Projects

- Developed a Super Mario Bros. agent using Double DQN with Python, PyTorch, and Gymnasium environment, achieving human-level performance of 30 seconds.
- Established an autonomous vehicle simulation, reduced 20% in lane deviation by applying transfer learning from obstacle detection to lane detection.

HONOURS AND AWARDS

Research Assistant Scholarship	September 2025 – August 2027
Academic Scholarships for Computer Science (Sangmyung University)	2019, 2022, 2023, 2024
Outstanding Program Performance (San Jose State University)	August 2024
Entrepreneurship Award (San Jose State University)	August 2024
Outstanding Computer Science Graduation Project Exhibition Award (Sangmyung University)	November 2024
Fourth Prize in Capstone Design Competition (Sangmyung University)	December 2024
Software Talent Scholarship (Sangmyung University; SW-Centered University Project Group)	December 2024
Army Warrior Award (Republic of Korea Army)	October 2021
The Army Commendation Medal: Outstanding Service (U.S. Army)	August 2021
Outstanding KATUSA Award (Republic of Korea Army)	May 2021
Excellence Award in Combined Command Post Training (U.S. Army)	September 2020

TECHNICAL SKILLS

- Programming Languages: Python, C, Java, C++, Rust, JavaScript, TypeScript, React
- Machine Learning Frameworks: PyTorch, Tensorflow, Keras, PyTorch Lightning
- Data Science: Pandas, NumPy, SQL
- Tools: Linux, Git, Docker, Anaconda, L^AT_EX

OTHERS

Military Service

April 2020 – October 2021

Korean Augmentation to the United States Army (KATUSA)

- Korea Army Squad Leader (Republic of Korea Army)